

Programme Name/s : Artificial Intelligence/ Artificial Intelligence and Machine Learning/ Data Sciences
Programme Code : AI/ AN/ DS
Semester : Fifth
Course Title : CLOUD COMPUTING FOR DATA SCIENCE
Course Code : 315327

I. RATIONALE

Cloud Computing play an important role for Data Science and Machine Learning Algorithm. It provides a flexible, scalable and cost-effective environment for data science which is important for computational needs, enhancing collaboration and ensuring robust data management with security. This Course will cover basic architecture of Cloud environment, use of various available cloud Services, build and develop machine learning algorithm to train and test the machine learning Model.

II. INDUSTRY / EMPLOYER EXPECTED OUTCOME

Use Cloud computing platform to solve real-world data science problems.

III. COURSE LEVEL LEARNING OUTCOMES (COS)

Students will be able to achieve & demonstrate the following COs on completion of course based learning

- CO1 - Use Cloud-Based software services by comprehending the cloud Computing Architecture.
- CO2 - Configure Virtual Machines using Virtualization techniques.
- CO3 - Implement Virtualized storage system in Cloud.
- CO4 - Use Machine Learning algorithms in Cloud Environment.
- CO5 - Deploy Machine Learning Models on Cloud.

IV. TEACHING-LEARNING & ASSESSMENT SCHEME

Course Code	Course Title	Abbr	Course Category/s	Learning Scheme						Credits	Assessment Scheme												Total Marks	
				Actual Contact Hrs./Week			SL	H	NL		Paper Duration	Theory				Based on LL & TL				Based on SL				
																Practical				SLA				
				CL	TL	LL						FA-TH	SA-TH	Total		FA-PR		SA-PR		SLA				
																Max	Min	Max	Min	Max	Min	Max		Min
315327	CLOUD COMPUTING FOR DATA SCIENCE	CCD	DSE	4	-	2	-	6	2	3	30	70	100	40	25	10	25#	10	-	-	150			

Total IKS Hrs for Sem. : 0 Hrs

Abbreviations: CL- ClassRoom Learning , TL- Tutorial Learning, LL-Laboratory Learning, SLH-Self Learning Hours, NLH-Notional Learning Hours, FA - Formative Assessment, SA -Summative assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment

Legends: @ Internal Assessment, # External Assessment, *# On Line Examination , @\$ Internal Online Examination

Note :

1. FA-TH represents average of two class tests of 30 marks each conducted during the semester.
2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester.
3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work.
4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 10 Weeks
5. 1 credit is equivalent to 30 Notional hrs.
6. * Self learning hours shall not be reflected in the Time Table.
7. * Self learning includes micro project / assignment / other activities.

V. THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
1	TLO 1.1 Explain the Evolution of Cloud Computing. TLO 1.2 Explain characteristics of Cloud Computing. TLO 1.3 Describe the Architecture of Cloud Computing.	Unit - I Introduction to Cloud Computing 1.1 Introduction to Cloud Computing: Definition, Evolution of Cloud computing (from Mainframes to Clouds), Service – Oriented Architecture, Web Services, Grid Computing, Utility Computing 1.2 Characteristics of a Cloud computing 1.3 Cloud computing architecture: Basic components: front-end platform, back-end, platform, Networking, cloud-based delivery 1.4 Cloud Service Models: Software as a Service (SaaS), Infrastructure as a Service (IaaS), Platform as a Service (PaaS), Continuous delivery using PaaS	Lecture Using Chalk-Board Flipped Classroom Presentations
2	TLO 2.1 Explain Concept of Virtualization. TLO 2.2 Explain the given type of Virtualization. TLO 2.3 Compare various cloud service models. TLO 2.4 Describe various Cloud Deployment Models. TLO 2.5 Explain role of Cloud computing in Data Science.	Unit - II Virtualization and Cloud Computing 2.1 Introduction to Virtualization, types of Virtualization, Application Virtualization, Network Virtualization, Desktop Virtualization, Storage Virtualization, Server Virtualization, Data virtualization 2.2 Cloud Deployment Models: Public, Private, Community, Hybrid 2.3 Role of Cloud computing in data Science 2.4 Advantages of Cloud Computing in Machine Learning	Lecture Using Chalk-Board Presentations Flipped Classroom

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
3	TLO 3.1 Explain Cloud Storage. TLO 3.2 Describe the given Cloud storage system. TLO 3.3 Explain features and limitations of Key-Value databases. TLO 3.4 Compare Batch data and Streaming data in Machine learning. TLO 3.5 Explain GCP BigQuery cloud platform.	Unit - III Cloud Storage 3.1 Cloud Storage: Introduction, Benefits of using Cloud Storage, Use cases of Cloud Storage (Backup, Archiving, Disaster recovery, Data processing, Content delivery) 3.2 Cloud storage system: Block-Based, File-Based, Object-Based Storages 3.3 Key-Value Databases: Introduction, features, limitations 3.4 Batch data and Streaming data in Machine learning 3.5 Cloud Data Warehouse– AWS Redshift 3.6 Various Cloud-based tools used for data science in ML– GCP BigQuery	Lecture Using Chalk-Board Presentations Demonstration
4	TLO 4.1 Explain benefits of using Machine Learning in the Cloud. TLO 4.2 Explain the given type of Cloud-Based Machine Learning Service. TLO 4.3 Describe various Machine Learning systems. TLO 4.4 Describe features of the given Cloud Machine Learning Platforms.	Unit - IV Cloud Computing for Data Science 4.1 Machine Learning in the Cloud: Benefits and Limitations 4.2 Types of Cloud-Based Machine Learning Services: Artificial Intelligence as a Service (AIaaS), GPU as a Service (GPUaaS) 4.3 Introduction to various ML systems and benefits of using Managed ML platforms 4.4 Cloud Machine Learning Platforms: AWS SageMaker, Azure Machine Learning studio, Google Cloud AutoML	Lecture Using Chalk-Board Presentations Demonstration
5	TLO 5.1 Explain factors for Selection of Cloud Machine Learning Platform. TLO 5.2 Describe ETL and ELT Pipelines. TLO 5.3 Explain Pre-Tuned AI Services. TLO 5.4 Write step wise procedure to train a Machine Learning Project.	Unit - V Training and Deployment of ML on Cloud 5.1 Factors for Selection of Cloud Machine Learning Platform, Support for ETL or ELT Pipelines, Support for Scale-Up and Scale-Out Training, Support for Machine Learning Frameworks, Pre-Tuned AI Services, Monitor Prediction Performance 5.2 Training Machine Learning Projects in the Cloud: Steps to train Machine Learning project in the Cloud, Identify and Understand Data Sources, Engineer the Features, Train and Validate Model, Deploy and Monitor Model	Lecture Using Chalk-Board Presentations Demonstration

VI. LABORATORY LEARNING OUTCOME AND ALIGNED PRACTICAL / TUTORIAL EXPERIENCES.

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 1.1 Use Google Cloud Software Services.	1	* Use Google Services to create Doc/Sheet/Keep/Forms	2	CO1
LLO 2.1 Create Virtual Machine.	2	*Create Virtual Machine using VMware workstation for Windows/Linux	2	CO2
LLO 3.1 Create Web Server on Virtual Machine.	3	Create Web server (WAMP/XAMP/APACHE) on Virtual Machine	2	CO2

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 4.1 Create account on any cloud platform.	4	*Create an account on AWS or Azure, or Google Cloud Platform	2	CO2
LLO 5.1 Use BigQuery to create tables and views.	5	Create an account on Google Cloud Platform. a) Create a project and access BigQuery. b) Query data directly from Google Sheets. c) Create tables and views using BigQuery	2	CO3
LLO 6.1 Create instance and configure Block-Based storage.	6	Launch an EC2 instance with a specified configuration and configure Block-Based storage	2	CO3
LLO 7.1 Configure File-Based storage.	7	*Create and Configure File-Based storage on EC2 instance	2	CO3
LLO 8.1 Configure Object-Based storage.	8	Create and Configure Object-Based storage on EC2 instance	2	CO3
LLO 9.1 Upload dataset to Aws S3 bucket using Python script.	9	Write a script in Python to upload a dataset to an S3 bucket, list the files in the bucket, and download a file	2	CO3
LLO 10.1 Use Amazon Sagemaker notebook.	10	*Create instance of Amazon Sagemaker notebook	2	CO4
LLO 11.1 Build and Deploy a model using Amazon Sagemaker.	11	* Build, test, tune, train, deploy and validate a model using Amazon Sagemaker	2	CO4 CO5

Note : Out of above suggestive LLOs -

- '*' Marked Practicals (LLOs) Are mandatory.
- Minimum 80% of above list of lab experiment are to be performed.
- Judicial mix of LLOs are to be performed to achieve desired outcomes.

VII. SUGGESTED MICRO PROJECT / ASSIGNMENT/ ACTIVITIES FOR SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)

Micro project

- Use AWS Glue, Google Dataflow, or Azure Data Factory to create a data pipeline that ingests, transforms, and loads data from one storage service to another. Document the pipeline configuration and execution results.
- Create a Machine on AWS and make basic services available on machine like word, power point etc.
- Set up a relational database using AWS RDS, Azure SQL Database, or Google Cloud SQL. Create a database schema, insert sample data, and perform queries using SQL. Document the steps and results.
- Launch an EC2 instance with a specified configuration (e.g., type, region). Install software (e.g., Python, Jupyter Notebook) on the instance. Connect to the instance and run a basic Python script.
- Create a local cloud on Virtual Machine on VmWare workstation. Deploy a web application and make it accessible by URL.
- Write and deploy a simple AWS Lambda function or Azure Function that processes data (e.g., transforms a JSON file). Test the function with sample data and set up triggers for automatic execution.

Note :

- Above is just a suggestive list of microprojects and assignments; faculty must prepare their own bank of microprojects, assignments, and activities in a similar way.
- The faculty must allocate judicious mix of tasks, considering the weaknesses and / strengths of the student in acquiring the desired skills.
- If a microproject is assigned, it is expected to be completed as a group activity.
- SLA marks shall be awarded as per the continuous assessment record.
- For courses with no SLA component the list of suggestive microprojects / assignments/ activities are optional, faculty may encourage students to perform these tasks for enhanced learning experiences.
- If the course does not have associated SLA component, above suggestive listings is applicable to Tutorials and maybe considered for FA-PR evaluations.

VIII. LABORATORY EQUIPMENT / INSTRUMENTS / TOOLS / SOFTWARE REQUIRED

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
1	VmWare Workstation Software	2,3
2	Aws, EC2 and Sagemaker instance with services	5,6,7,8,9,10,11
3	Minimum Hardware requirement: Personal computer, (i3-i7 preferable), RAM minimum 8 GB.	All
4	Minimum system requirement: 64-bit operating system such as Windows 10, macOS 10.13, or a recent version of Linux.	All

IX. SUGGESTED WEIGHTAGE TO LEARNING EFFORTS & ASSESSMENT PURPOSE (Specification Table)

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	Introduction to Cloud Computing	CO1	6	4	8	0	12
2	II	Virtualization and Cloud Computing	CO2	8	4	6	4	14
3	III	Cloud Storage	CO3	8	4	6	4	14
4	IV	Cloud Computing for Data Science	CO4	10	4	8	4	16
5	V	Training and Deployment of ML on Cloud	CO5	8	4	6	4	14
Grand Total				40	20	34	16	70

X. ASSESSMENT METHODOLOGIES/TOOLS**Formative assessment (Assessment for Learning)**

- Laboratory Performance, Unit Tests , Midterm Exam, Term Work, Seminar/Presentations.
- Continuous assessment based on process and product related performance indicators
- Each practical will be assessed considering 60% weightage to process and 40% weightage to product

Summative Assessment (Assessment of Learning)

- End Semester Exam, Practical exam, viva voce.

XI. SUGGESTED COS - POS MATRIX FORM

Course Outcomes (COs)	Programme Outcomes (POs)							Programme Specific Outcomes* (PSOs)		
	PO-1 Basic and Discipline Specific Knowledge	PO-2 Problem Analysis	PO-3 Design/ Development of Solutions	PO-4 Engineering Tools	PO-5 Engineering Practices for Society, Sustainability and Environment	PO-6 Project Management	PO-7 Life Long Learning	PSO-1	PSO-2	PSO-3
CO1	2	2	2	1	2	1	1			
CO2	2	3	2	3	3	1	1			
CO3	2	2	2	2	2	1	1			
CO4	3	2	3	3	2	1	1			
CO5	3	2	3	3	2	2	2			
Legends :- High:03, Medium:02,Low:01, No Mapping: - *PSOs are to be formulated at institute level										

XII. SUGGESTED LEARNING MATERIALS / BOOKS

Sr.No	Author	Title	Publisher with ISBN Number
1	Dr. Anand Nayyar	Handbook of Cloud Computing	BPB Publication First Edition (1 January 2019) ISBN-10:9388176669 ISBN-13:978-9388176668
2	Toby Velte, Anthony Velte, Robert C	Cloud Computing: A Practical Approach By Toby Velte, Anthony Velte, Robert C	McGraw Hill Professional ISBN-978-0-07-162965-8
3	Noah Gift, Alfredo Deza	Cloud Computing for Data Analysis	Pragmatic AI Labs (No ISBN)
4	Valliappa Lakshmanan	Data Science on the Google Cloud Platform: Implementing End-to-End Real-Time Data Pipelines	O'Reilly Media, Inc. ISBN: 9781098118952
5	Abhishek Mishra	Machine Learning in the AWS Cloud: Add Intelligence to Applications with Amazon SageMaker	Wiley Publication ISBN: 978-1-119-55671-8

XIII. LEARNING WEBSITES & PORTALS

Sr.No	Link / Portal	Description
1	https://www.geeksforgeeks.org/virtualization-cloud-computing-types/	Introduction to virtualization and cloud Computing and its types
2	https://www.javatpoint.com/virtualization-in-cloud-computing	Introduction to virtualization and cloud Computing
3	https://www.hostitsmart.com/blog/types-of-virtualization-in-cloud-computing-complete-overview/	Overview of cloud computing, types of Virtualization advantages and Application
4	https://aws.amazon.com/what-is/virtualization/	How can AWS help with virtualization and cloud computing?
5	https://www.run.ai/guides/machine-learning-in-the-cloud	Machine Learning in the Cloud,AWS Sagemaker Service for Machine Learning

Sr.No	Link / Portal	Description
Note : Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students		